

Filtering Data in SQL Lab

Summary: Applying filters to SQL queries helps to retrieve specific information from an organization's database. In this scenario, I was tasked with identifying which employees' devices required security updates and locating employees in key departments. The goal of this lab was to further understand how to query specific information with SQL in the most efficient way.

Commands Used:



Overview: First, I started by retrieving a list of all the machines in the organization by using a simple `SELECT` statement. This gave me an overview of the device IDs and their operating systems. After reviewing the full list, I applied the `WHERE` filter to search for machines running on OS 2, since those devices required updates. This clause returns only the rows that match a specific condition, such as `WHERE operating_system = 'OS 2'`. Filtering the results made it easier to focus on the systems that needed attention.

I then moved on to examine employee information to find those in a particular department to send an updated policy notice. I used `WHERE` again to return only the rows where the department matched the value I needed. These filtered results helped identify which employees needed to receive the notice.

In the final task, I used filters to identify which offices these employees were working in. To do this, I used the `LIKE` operator, which searches for patterns in text. This operation can be combined with the wildcard `%`, which represents any number of characters. For example, using `LIKE 'South%'` returns all offices that begin with "South," such as South-145. This pattern matching made it easy to locate all employees working in the South building who needed to receive the updated notices.

Reflection: Using filtering tools makes it much easier to isolate specific devices, departments, and office locations without scanning entire tables. These filtering techniques are essential for security analysts who need to quickly identify issues and focus on the most relevant information during an investigation. This lab highlighted how SQL can narrow large datasets into clear, actionable results, which is critical when working with real organizational data.